

UNIT 2

Water Pollution & Its Control Techniques

Environmental Science | Detailed Study Notes

■ 6 Credits

■ 7-Mark Questions Included

■ Full Syllabus Covered

1. Introduction to Water Pollution

Water is the most precious natural resource on Earth. However, human activities, industrialisation, agricultural practices, and urbanisation have severely degraded water quality worldwide. **Water pollution** is defined as the introduction of undesirable substances — physical, chemical, or biological — into a water body that renders it unfit for its intended use (drinking, irrigation, aquatic life, recreation).

Definition: Water pollution occurs when harmful substances (pollutants) contaminate streams, rivers, lakes, oceans, groundwater, or other water bodies, degrading water quality and making it toxic for humans and the environment.

Types of Water Pollutants:

- **Physical pollutants:** Suspended solids, colour, turbidity, heat (thermal pollution)
- **Chemical pollutants:** Heavy metals, pesticides, fertilizers, acids, organic compounds
- **Biological pollutants:** Pathogens — bacteria, viruses, protozoa, parasites
- **Radiological pollutants:** Radioactive isotopes from nuclear plants or mining

2. Major Sources of Water Pollution

2.1 Eutrophication

Eutrophication is the process by which a water body becomes overly enriched with nutrients (primarily nitrogen and phosphorus), stimulating excessive growth of algae and aquatic plants. It is one of the most serious water quality problems globally.

Causes of Eutrophication:

- Agricultural run-off carrying nitrogen (NO_3^-) and phosphate (PO_4^{3-}) from fertilizers
- Discharge of municipal sewage and detergents rich in phosphorus
- Animal wastes from livestock farming entering water bodies
- Industrial effluents containing organic matter
- Urban storm-water run-off

Process and Effects:

- Excess nutrients → rapid algal growth → **Algal Bloom** (green/blue-green scum on water surface)
- Algal bloom blocks sunlight → aquatic plants below die
- Dead organic matter decomposed by bacteria → **Biological Oxygen Demand (BOD)** rises sharply
- **Hypoxia / Anoxia:** Dissolved oxygen (DO) drops below 2 mg/L → fish kills
- Release of toxins by cyanobacteria (blue-green algae) — harmful to humans and animals

- Loss of biodiversity; lake/pond eventually becomes a "dead zone"

Key Terms: Oligotrophic (nutrient-poor, clear) → Mesotrophic → Eutrophic → Hypertrophic (severely polluted). Cultural eutrophication is human-accelerated eutrophication.

2.2 Acid Mine Drainage (AMD)

Acid Mine Drainage (AMD) is formed when iron sulphide minerals (especially pyrite — FeS_2) in rock, soil, or mine tailings are exposed to air and water. Chemical oxidation and microbial activity produce sulphuric acid, which leaches into streams, rivers, and groundwater.

Chemical Reaction (Pyrite Oxidation):

$$\text{FeS}_2 + \text{H}_2\text{O} + \text{O}_2 \rightarrow \text{Fe}^{2+} + \text{SO}_4^{2-} + \text{H}^+ \text{ (Acid)}$$

Fe^{2+} is further oxidised to Fe^{3+} (orange/red precipitate — "yellow boy") by bacteria like *Acidithiobacillus ferrooxidans*.

Effects of AMD:

- pH of water drops to 2–4 (extremely acidic) — lethal to most aquatic organisms
- Mobilises heavy metals: arsenic (As), lead (Pb), cadmium (Cd), mercury (Hg), zinc (Zn)
- Orange/red staining of stream beds; characteristic metallic odour
- Destruction of benthic (bottom-dwelling) organisms and fish populations
- Contamination of drinking water sources and agricultural land
- Persists for decades or centuries even after mine closure

Control Measures for AMD:

- **Active treatment:** Lime dosing to neutralise acid; chemical precipitation of metals
- **Passive treatment:** Constructed wetlands, anoxic limestone drains, permeable reactive barriers
- **Prevention:** Backfilling mines, surface sealing to exclude oxygen and water

2.3 Pesticides and Fertilizers

Modern agriculture relies heavily on synthetic pesticides and fertilizers. When applied in excess or washed by rain, these chemicals enter water bodies causing serious pollution.

Pesticide Pollution:

- Categories: Organochlorines (DDT, Aldrin, Endrin), Organophosphates (Malathion, Parathion), Carbamates, Pyrethroids
- **Bioaccumulation:** Pesticides are fat-soluble; concentrated in body fat of organisms
- **Biomagnification:** Concentration increases at each trophic level (up to 1000× in top predators)

- DDT → eggshell thinning in birds; disrupts endocrine system; carcinogenic in mammals
- Runoff of pesticides → kills non-target aquatic insects, crustaceans, and fish
- Leaching to groundwater renders wells unfit for drinking

Fertilizer Pollution:

- **Nitrate leaching:** NO_3^- enters groundwater; causes **Methemoglobinemia (Blue Baby Syndrome)** in infants
- Excess phosphate → eutrophication (as discussed above)
- Ammonia from urea fertilizers is directly toxic to fish at low concentrations
- Fluoride impurities in phosphate fertilizers accumulate in soil and water

2.4 Dyeing and Tanning Industries

The textile dyeing and leather tanning industries are among the most polluting industries because of the large volumes of chemically-loaded effluents they discharge.

Dyeing Industry Pollutants:

- Synthetic dyes: Azo dyes, reactive dyes, vat dyes — highly coloured, toxic, and resistant to biodegradation
- Auxiliaries: Salts (NaCl), surfactants, fixing agents, heavy metals (chromium in mordant dyeing)
- Effluent characteristics: High BOD, COD, TDS, colour, pH variations
- Azo dye degradation produces **aromatic amines** — potent carcinogens
- Coloured water blocks sunlight penetration → affects photosynthesis in aquatic plants

Tanning Industry Pollutants:

- **Chrome tanning (most common):** Uses hexavalent chromium Cr(VI) — carcinogenic and mutagenic
- Effluent contains: sulphides, lime, salts, ammonium compounds, organic nitrogen
- High suspended solids, BOD (800–3000 mg/L), TDS, and strong odour
- **Chromium toxicity:** Damages kidneys, liver; causes skin ulcers and lung cancer
- Untreated tannery effluent destroys aquatic biodiversity completely

2. Major Sources (continued)

2.5 Marine Pollution

The oceans cover 71% of Earth's surface and are the ultimate sink for most water pollutants. Marine pollution threatens marine biodiversity, fisheries, and coastal communities worldwide.

Major Sources of Marine Pollution:

- **Oil spills:** Tanker accidents, offshore drilling blow-outs, pipeline leaks — form slicks that coat marine organisms
- **Plastic debris:** ~ 8 million tonnes of plastic enter oceans annually
- **Heavy metals:** Mercury, lead, cadmium — bioaccumulate in marine food chains
- **Sewage and nutrients:** Cause coastal eutrophication and "dead zones" (e.g., Gulf of Mexico)
- **Radioactive waste:** Historical dumping of nuclear waste containers
- **Noise pollution:** Shipping and sonar affect whale communication and navigation
- **Thermal pollution:** Cooling water from coastal power plants raises sea temperature

Effects of Oil Spills:

- Smothers marine birds and mammals; impairs thermoregulation by coating feathers/fur
- Toxic PAHs (polycyclic aromatic hydrocarbons) damage gills, livers, and reproduction of fish
- Coral reef bleaching and mortality
- Destroys mangroves and intertidal ecosystems
- Economic losses to fishing and tourism industries

2.6 Micro-plastics

Micro-plastics are plastic particles **smaller than 5 mm** in size. They have emerged as one of the most pervasive and alarming pollutants of the 21st century, found everywhere from ocean trenches to Arctic ice to human blood.

Types and Sources:

- **Primary micro-plastics:** Manufactured directly at micro-scale — microbeads in cosmetics/personal care products, plastic pellets (nurdles)
- **Secondary micro-plastics:** Result from breakdown of larger plastic items through UV, mechanical abrasion, and weathering
- **Synthetic fibres:** Released during washing of polyester/nylon clothing (up to 700,000 fibres per wash)
- **Tyre wear particles:** Major source — enter stormwater drains

- **Agricultural films:** Degrading plastic mulch films fragment in soil

Effects of Micro-plastics:

- Ingested by fish, seabirds, marine mammals, and filter-feeders (mussels, oysters)
- Physical blockage of digestive systems; false sense of satiety → starvation
- Carry and concentrate persistent organic pollutants (PCBs, DDT) on their surface — "Trojan Horse effect"
- Enter human food chain via seafood, drinking water, and even table salt
- Nano-plastics (<100 nm) can cross cell membranes and blood-brain barrier
- Endocrine disruption due to plastic additives (phthalates, bisphenol-A)

Did you know? A recent study found micro-plastics in human placenta, breast milk, and lungs. The average person ingests approximately 5 grams of plastic per week — equivalent to a credit card.

3. Techniques to Control Water Pollution

Water pollution control involves conventional and advanced treatment technologies to remove pollutants from wastewater before it is discharged into the environment or reused.

3.1 Conventional Wastewater Treatment — Types of Sewage

Sewage is the liquid waste from households, commercial establishments, and some industries. It contains 99.9% water and 0.1% impurities, yet this small fraction causes enormous environmental damage.

Types of Sewage:

- **Domestic/Sanitary sewage:** From households — contains organic matter, pathogens, nutrients (N, P), detergents
- **Industrial sewage/effluents:** From factories — may contain heavy metals, toxic chemicals, high BOD/COD
- **Stormwater/Surface runoff:** Rain-driven runoff from roads, roofs — carries silt, oil, pesticides
- **Combined sewage:** Mixture of domestic and stormwater in older piped systems

3.2 Sewerage System

A **sewerage system** is the network of pipes, pumping stations, and infrastructure that collects and transports sewage from its source to treatment facilities.

Components of Sewerage System:

- **House sewer:** Connects individual buildings to the public sewer network
- **Lateral/branch sewers:** Smaller pipes collecting from groups of buildings
- **Main sewers/trunk sewers:** Large pipes carrying sewage to interceptor sewers
- **Interceptor sewers:** Large conduits running parallel to rivers, intercepting outfalls
- **Pumping/lift stations:** Raise sewage to higher elevations when gravity flow is insufficient
- **Sewage treatment plant (STP):** Final destination for treatment

3.3 Alternative Systems

- **Septic tanks:** On-site treatment for areas not connected to sewers — anaerobic digestion of solids
- **Constructed wetlands:** Engineered wetland systems using plants and microbes — low cost, eco-friendly
- **Waste stabilisation ponds:** Large shallow lagoons using sunlight and algae for treatment
- **Package treatment plants:** Compact, pre-fabricated units for small communities or resorts

- **Biodigesters / biogas plants:** Anaerobic treatment producing biogas (CH_4) as energy

4. Wastewater Treatment — Primary, Secondary & Tertiary

4.1 Primary Treatment (Physical Treatment)

Primary treatment uses physical processes to remove large and settleable solids from wastewater. It typically removes 50–65% of suspended solids and 30–40% of BOD.

Unit Operations in Primary Treatment:

- **Screening:** Bar screens or fine screens remove large objects (rags, sticks, plastics). Coarse screens: 50–150 mm openings; Fine screens: 3–6 mm
- **Grit removal:** Grit chambers slow velocity to 0.3 m/s; sand, gravel, and eggshells settle
- **Primary sedimentation (clarifiers):** Wastewater held 1.5–2 hours; organic solids settle as **primary sludge**; grease and oil float and are skimmed off
- **Flotation:** Dissolved air flotation (DAF) for light solids and fats
- **Equalization:** Balances flow rates and pollutant loads before further treatment

Primary Sludge collected from clarifiers is typically 3–8% solids. It must be further processed through sludge digestion, thickening, dewatering, and disposal.

4.2 Secondary Treatment (Biological Treatment)

Secondary treatment uses biological processes — aerobic or anaerobic — to remove dissolved and colloidal organic matter. It reduces BOD by 85–95% and removes 90% of suspended solids.

Aerobic Processes:

- **Activated Sludge Process (ASP):** Wastewater mixed with micro-organism-rich "activated sludge" in aeration tanks. Air blown in; bacteria oxidise organic matter. Effluent clarified; sludge recycled. Most widely used secondary treatment worldwide.
- **Trickling Filter (Biofilter):** Wastewater sprinkled over a bed of stones/plastic media. Biofilm of bacteria on media degrades organics. Simple, low energy, but less efficient than ASP.
- **Rotating Biological Contactor (RBC):** Plastic discs half-submerged, rotate slowly. Biofilm alternately exposed to wastewater and air. Compact and energy-efficient.
- **Oxidation Ponds / Lagoons:** Shallow basins (0.3–1.5 m deep); sunlight, wind, and algae support aerobic bacteria. Low-cost, suitable for tropical countries.
- **Sequential Batch Reactor (SBR):** Fill–React–Settle–Decant cycle in a single tank. Flexible and suitable for small flows.

Anaerobic Processes:

Anaerobic treatment occurs **without oxygen**. Used for high-strength industrial wastewater and sludge treatment. Produces biogas (60–70% CH₄) as a valuable by-product.

- **Anaerobic Digestion:** Four stages — (1) **Hydrolysis:** complex organics broken to monomers; (2) **Acidogenesis:** monomers → volatile fatty acids + H₂ + CO₂; (3) **Acetogenesis:** fatty acids → acetate; (4) **Methanogenesis:** acetate + H₂ → CH₄ + CO₂
- **Upflow Anaerobic Sludge Blanket (UASB) Reactor:** Wastewater flows upward through a sludge blanket. Very compact; widely used in India for textile and sugar factory effluents.
- **Anaerobic Filter:** Fixed-film anaerobic reactor with inert media for biofilm attachment.
- **Imhoff Tank:** Two-chamber tank — upper settling compartment + lower digestion compartment. Traditional design, suitable for small communities.

Parameter	Aerobic Treatment	Anaerobic Treatment
Oxygen requirement	Requires oxygen (aeration)	No oxygen — strict anaerobes
Energy	Energy intensive (aeration costs)	Low energy; produces biogas
BOD removal	85–95%	50–85%
Sludge produced	High quantity	Low quantity
By-products	CO ₂ , water, cell biomass	Biogas (CH ₄ + CO ₂), digestate
Start-up time	1–2 weeks	2–4 months
Application	Municipal sewage, low-strength waste	High-strength industrial waste

4.3 Tertiary Treatment (Advanced Treatment)

Tertiary treatment further polishes the secondary effluent to produce high-quality water suitable for reuse, discharge into sensitive water bodies, or irrigation. It can remove remaining nutrients, pathogens, and trace contaminants.

- **Nutrient removal (Nitrogen):** Nitrification ($\text{NH}_4^+ \rightarrow \text{NO}_3^-$) followed by Denitrification ($\text{NO}_3^- \rightarrow \text{N}_2$ gas) using bacteria in anoxic zones
- **Phosphorus removal:** Chemical precipitation using alum ($\text{Al}_2(\text{SO}_4)_3$) or lime; or biological enhanced phosphorus removal (EBPR) in activated sludge
- **Filtration:** Rapid sand filters, multimedia filters, or membrane filters (MF, UF, NF, RO) to remove fine suspended particles and pathogens
- **Disinfection:** Chlorination (most common), UV irradiation (chemical-free), ozonation (powerful oxidiser), or chloramine treatment for pathogen destruction
- **Activated carbon adsorption:** Granular activated carbon (GAC) removes trace organics, pharmaceuticals, pesticides, and colour
- **Reverse osmosis (RO):** Semi-permeable membranes remove dissolved salts, heavy metals, nitrates — used for water reuse and desalination
- **Advanced oxidation processes (AOPs):** H_2O_2 /UV, Fenton's reagent — destroy recalcitrant micro-pollutants

Reclaimed Water Quality: After tertiary treatment, water meets standards for agricultural irrigation (Class A), industrial cooling, groundwater recharge, and in some countries, indirect potable reuse.

4.4 Safe Disposal and Its Utility

Proper disposal of treated wastewater and sludge is critical to prevent secondary pollution.

Methods of Safe Disposal:

- **Discharge to surface water:** Treated effluent must meet CPCB/SPCB standards (BOD <30 mg/L, SS <100 mg/L, pH 5.5–9)
- **Land disposal (irrigation):** Treated wastewater used for agriculture (rich in N, P); soil acts as additional treatment medium — "living filter"
- **Sludge utilization:** Dried/composted sludge (biosolids) used as soil amendment/fertilizer
- **Biogas utilization:** Methane from digesters used for electricity, cooking, or heat
- **Effluent reuse:** In industries for cooling towers, boilers (water conservation)
- **Ocean outfall:** Treated effluent discharged via submarine outfall pipes — requires EIA approval

5. Treatment Schemes for Industrial Wastewater

5.1 Dairy Industry Wastewater

Dairy wastewater originates from milk processing, cheese and butter making, yoghurt production, and cleaning of equipment. It has **very high BOD (1,000–5,000 mg/L)** due to milk fats, proteins, and lactose.

Characteristics:

- High BOD and COD from milk solids; milky/white appearance
- High fat content (emulsified milk fat)
- Fluctuating pH from acid cleaning (pH 2) and alkaline CIP (pH 12)
- High temperature effluent from pasteurisation
- Contains detergents and sanitisers (chlorine compounds)

Treatment Scheme:

- **Step 1 — Screening:** Remove coarse solids
- **Step 2 — Equalization tank:** Balance flow and pH fluctuations
- **Step 3 — DAF (Dissolved Air Flotation):** Remove emulsified fats and suspended solids
- **Step 4 — Aerobic biological treatment:** Activated sludge or trickling filter for BOD reduction
- **Step 5 — Secondary clarifier:** Separate biomass from treated effluent
- **Step 6 — Anaerobic digestion of sludge:** Produce biogas; recover energy
- **Step 7 — Polishing / Disinfection:** Chlorination before discharge

5.2 Textile Industry Wastewater

Textile effluents are among the most complex and environmentally damaging industrial wastewaters. They are **highly coloured** (from dyes), alkaline, and contain a wide array of toxic chemicals.

Characteristics:

- Strong colour (absorbs light) — impairs aquatic photosynthesis even at very low concentrations
- High BOD (200–600 mg/L), very high COD (400–3000 mg/L)
- High TDS and conductivity (due to salts used in dyeing)
- Alkaline pH (8–12); high temperature
- Heavy metals from metallic dyes (chromium, copper)
- Toxic surfactants and auxiliary chemicals

Treatment Scheme:

- **Primary:** Screening → Equalization → Coagulation-flocculation (alum/ FeCl_3) → Sedimentation — removes colour and SS
- **Secondary:** Biological treatment — ASP or UASB + aerobic (for high BOD reduction)
- **Tertiary:** Activated carbon adsorption (for residual colour and organics) → Ozonation/UV (colour destruction) → Reverse osmosis (for zero liquid discharge)
- **ZLD (Zero Liquid Discharge):** Many Indian states mandate ZLD for textile mills — all water recovered and recycled; only solid waste remains

5.3 Power Plant Wastewater

Thermal power plants (coal, gas, nuclear) generate multiple streams of wastewater from cooling, ash handling, flue gas desulfurization, and boiler operations.

- **Cooling water:** High temperature (thermal pollution); biocides (Cl_2) added to prevent biological fouling
- **Ash pond water:** Contains fly ash slurry — high suspended solids, heavy metals (As, Hg, Se), fluoride
- **FGD wastewater:** Flue gas desulphurisation effluent — high selenium, mercury, nitrates
- **Boiler blowdown:** High TDS, dissolved minerals, traces of hydrazine and morpholine

Treatment:

- Cooling water: Cooling towers (reduce temperature); biocide treatment; drift eliminator
- Ash pond: Settling basins → coagulation → sludge dewatering; fly ash utilised in cement/bricks
- FGD water: Chemical precipitation (lime) → membrane treatment → evaporation for ZLD

5.4 Pharmaceutical Industry Wastewater

Pharmaceutical effluents are characterised by the presence of **Active Pharmaceutical Ingredients (APIs), solvents, and intermediates** that are toxic, recalcitrant (not easily biodegraded), and potentially carcinogenic or mutagenic.

- Very high BOD (up to 50,000 mg/L for some fermentation streams) and COD
- Presence of antibiotics — inhibit biological treatment bacteria if untreated
- Residual antibiotics in water bodies promote antibiotic-resistant bacteria (AMR — major global health threat)
- Solvents: methanol, acetone, chlorinated solvents — toxic and volatile
- Heavy metals from catalyst and reagent usage

Treatment Scheme:

- **Solvent recovery:** Distillation to recover and reuse solvents before wastewater treatment
- **Physico-chemical treatment:** pH adjustment → coagulation-flocculation → sedimentation

- **Advanced oxidation (AOP):** Ozone, H_2O_2 /UV, or Fenton process to destroy APIs and solvents
- **Biological treatment:** High-rate anaerobic (UASB) + aerobic (ASP) — only after AOP pre-treatment
- **Activated carbon polishing:** Remove residual micro-pollutants
- **RO + Evaporation:** For ZLD compliance

5.5 Agro-Based Industries — Rice Mills

Rice mills produce wastewater primarily from parboiling (soaking and steaming of paddy before milling). This is called **parboiling effluent** and is one of the most strongly polluting agro-industrial wastewaters.

Characteristics of Rice Mill Wastewater:

- Extremely high BOD (5,000–12,000 mg/L) and COD — due to starch, proteins, sugars from rice leaching
- Dark brown colour from parboiling
- High suspended solids (rice husk fragments, bran particles)
- Strong odour from anaerobic decomposition
- Slightly acidic pH (5–6.5)

Treatment Scheme:

- **Screening:** Remove rice husk and fibrous solids
- **Settling / Primary clarifier:** Remove heavy suspended solids (starch granules)
- **Anaerobic digestion (primary biological):** UASB reactor or covered lagoon — most suitable for this high-BOD waste; produces biogas used for parboiling steam generation
- **Aerobic polishing:** Activated sludge or oxidation pond for remaining BOD
- **Constructed wetlands:** Final polishing — low-cost, eco-friendly
- **Resource recovery:** Rice husk → silica / ash for cement; biogas → energy; rice bran → oil extraction

Zero-Waste Approach: Modern rice mills aim for a zero-waste concept: parboiling effluent → biogas (energy) → digestate (fertilizer); rice husk → fuel + silica; rice bran → edible oil.

6. Important 7-Mark / Long-Answer Questions

Exam Tip: 7-mark answers should be approximately 300–500 words. Include definitions, mechanisms, a diagram description (if applicable), examples, and at least one comparison table where relevant.

Q1. What is eutrophication? Explain its causes, process, and effects on aquatic ecosystems with a suitable diagram.

Introduction: Eutrophication is the process by which a water body becomes enriched with inorganic nutrients — especially nitrogen (N) and phosphorus (P) — leading to excessive growth of algae and aquatic plants. It is a natural process but human activities (cultural eutrophication) have dramatically accelerated it.

Causes: (1) Agricultural run-off containing nitrate and phosphate fertilizers. (2) Discharge of municipal sewage and phosphate-rich detergents. (3) Animal waste from livestock entering surface water. (4) Industrial effluents containing organic nitrogen and phosphorus. (5) Urban storm-water carrying organic material from streets and lawns.

Process: The sequence is as follows: *Nutrient input* → *Algal bloom formation* (rapid growth of surface algae/cyanobacteria) → *Sunlight blockage* (sub-surface plants die) → *Decomposition by bacteria* (huge oxygen demand) → *Dissolved oxygen (DO) depletion (Hypoxia)* → *Fish kills and biodiversity loss*.

Effects: (a) Depletion of dissolved oxygen below 2 mg/L — fish suffocation. (b) Release of toxins by cyanobacteria (microcystin) — harmful to humans and livestock. (c) Foul odour and unaesthetic appearance — loss of recreational value. (d) Shift to algae-dominated ecosystem; loss of submerged aquatic vegetation. (e) Economic losses — reduced fish catch, contaminated drinking water sources. (f) Long-term: the water body may become shallow and eventually turns into a swampy land.

Control Measures: Reduce nutrient inputs by adopting precision fertilization; upgrading wastewater treatment plants to include nutrient removal (nitrification-denitrification for N; chemical phosphorus precipitation with alum); banning phosphate-based detergents; restoring riparian buffer strips along waterways.

Q2. Describe in detail the primary, secondary, and tertiary stages of wastewater treatment with appropriate diagrams and the significance of each stage.

Introduction: Municipal wastewater treatment is carried out in three progressively advanced stages, each targeting specific classes of pollutants.

Primary Treatment (Physical): Employs physical operations. (1) Screening: bar screens remove large debris; (2) Grit chamber: removes sand and gravel at controlled flow velocity (0.3 m/s); (3) Primary settling tank (clarifier): detention time 1.5–2 hours — 50–65% suspended solids settle as primary sludge; floating grease/oil is mechanically skimmed. Efficiency: removes 30–40% BOD and 50–65% SS.

Secondary Treatment (Biological): Uses microorganisms to oxidise dissolved organics. Aerobic processes: Activated Sludge Process (ASP) — aeration tank + secondary clarifier; Trickling filter — biofilm on media; Rotating biological contactor (RBC). Anaerobic processes: UASB reactor, anaerobic filter — for high-strength wastes. Efficiency: removes 85–95% BOD and 90% SS. Secondary sludge is generated.

Tertiary Treatment (Advanced): Polishes secondary effluent. Processes: Nutrient removal (nitrification + denitrification; phosphorus precipitation); Filtration (sand filters, membrane filtration); Disinfection (chlorination, UV, ozone); Activated carbon adsorption; Reverse osmosis. Produces near-potable quality water suitable for irrigation, industrial reuse, or groundwater recharge.

Significance: A well-designed three-stage system can reduce BOD from ~300 mg/L (raw sewage) to <5 mg/L (tertiary treated), meeting stringent standards. It protects receiving water bodies, prevents eutrophication, eliminates pathogens, and increasingly enables water recycling in water-scarce regions.

Q3. Write detailed notes on the treatment of textile industry wastewater. Why is Zero Liquid Discharge (ZLD) important?

Characteristics of Textile Effluent: Textile wastewater contains synthetic dyes (azo, reactive, vat dyes), large quantities of sodium chloride (salt used in dyeing), surfactants, alkaline pH (8–12), high temperature, high BOD (200–600 mg/L), very high COD (400–3000 mg/L), and heavy metals from metallic dyes. Azo dyes breakdown to produce aromatic amines — carcinogenic.

Treatment Scheme: Stage 1 — *Primary treatment*: Screening removes fibre and lint; equalization tank buffers pH and flow; coagulation-flocculation with alum or ferric chloride removes colour and suspended solids; sedimentation in clarifier. Stage 2 — *Secondary biological treatment*: ASP or UASB + aerobic system reduces BOD/COD. Note: most synthetic dyes are poorly biodegradable, so biological treatment alone is insufficient. Stage 3 — *Tertiary treatment*: Activated carbon adsorption removes residual colour and trace organics; ozonation or UV treatment destroys chromophore groups of dye molecules (decolourisation); Nanofiltration or Reverse Osmosis for desalination and water recovery.

Zero Liquid Discharge (ZLD): ZLD is a water management strategy where no liquid effluent is discharged outside the plant boundary. All treated water is recycled back into the process. Residuals are concentrated to solid/semi-solid form using evaporators and crystallisers. India's National Green Tribunal (NGT) and several State Pollution Control Boards have made ZLD mandatory for textile units in critically polluted areas.

Benefits of ZLD: (a) Zero water pollution of nearby rivers and groundwater. (b) Conservation of freshwater — recovered water reused in dyeing process. (c) Recovery of salt (NaCl) — economic benefit. (d) Compliance with environmental regulations. **Challenges:** High capital and operating cost; high energy consumption of evaporators; management of solid crystallised residue.

Q4. Explain in detail the sources, effects, and control measures for micro-plastic pollution in water bodies.

Introduction: Micro-plastics are plastic particles less than 5 mm in size. They have been detected in oceans, rivers, groundwater, drinking water, food, and even human blood and organs, representing a pervasive new class of environmental pollutants.

Sources: (1) Primary sources — manufactured at micro-scale: microbeads in cosmetics, plastic pellets (nurdles — raw material for plastic manufacturing). (2) Secondary sources — fragmentation of larger plastics by UV radiation, wave action, and mechanical abrasion: plastic bags, bottles, packaging. (3) Synthetic textile fibres released during laundry (up to 700,000 fibres per wash cycle). (4) Tyre wear particles — billions of particles enter stormwater drains. (5) Degrading plastic films used in agriculture.

Pathways into Water: Stormwater runoff → rivers → oceans; WWTP effluent (conventional treatment removes only 70–80% of micro-plastics); atmospheric deposition.

Effects: (a) Marine organisms ingest micro-plastics, causing gut blockage, false satiety, and reduced feeding. (b) Micro-plastics are vectors for POPs (persistent organic pollutants) — "Trojan horse effect." (c) Nano-plastics (<100 nm) penetrate cell membranes, blood-brain barrier, and placenta. (d) Plastic additives (BPA, phthalates) cause endocrine disruption. (e) Antibiotic resistance genes (ARGs) accumulate on micro-plastic surfaces. (f) Humans ingest approximately 5 g plastic/week through food, water, and air.

Control: Ban on microbeads in cosmetics (implemented in USA, UK, Canada, India). Extended Producer Responsibility (EPR) for plastic packaging. Improved wastewater treatment (fine filtration, membrane bioreactors) to capture micro-plastics. Better waste management to prevent plastic entering environment. Consumer awareness and reduction of single-use plastics.

Q5. Describe acid mine drainage — its formation, chemical reactions involved, environmental impacts, and treatment.

Definition: Acid Mine Drainage (AMD) is acidic, metal-rich water formed when iron sulphide minerals (especially pyrite, FeS_2) are exposed to air and water during and after mining operations. It is considered one of the most serious and long-lasting forms of water pollution globally.

Formation Mechanism: Pyrite oxidation proceeds in stages: (1) Chemical oxidation of pyrite by O_2 and water produces ferrous iron (Fe^{2+}) and sulphuric acid. (2) Fe^{2+} is oxidised to Fe^{3+} (ferric iron) — this step greatly accelerated by the bacterium *Acidithiobacillus ferrooxidans*. (3) Fe^{3+} oxidises more pyrite in a self-propagating cycle — the reaction becomes autocatalytic. (4) At low pH, heavy metals (Cu, Zn, Pb, Cd, As, Hg) dissolve from surrounding rock.

Characteristic Features: pH 2–4; bright orange/red precipitate of iron hydroxide ("yellow boy" or "orange slime") coating stream beds; metallic taste; elevated heavy metal concentrations; lifeless water (no fish or invertebrates).

Environmental Impacts: (a) Kills all aquatic life due to extreme acidity and metal toxicity. (b) Contaminates drinking water wells near mines. (c) Destroys agricultural soils if AMD-affected water is used for irrigation. (d) Infrastructure corrosion — pipelines, concrete structures. (e) Persists for centuries after mine closure — legacy pollution.

Treatment: Active — Lime neutralization raises pH to 8–9; metals precipitate as hydroxides and are filtered. Passive — Anoxic limestone drains (ALD); constructed wetlands (plants and sulphate-reducing bacteria neutralise acid and precipitate metals); permeable reactive barriers. Prevention — Mine sealing, backfilling, oxygen-barrier covers over tailings piles.

Q6. Compare aerobic and anaerobic wastewater treatment processes. Explain the stages of anaerobic digestion and its advantages.

Overview: Biological wastewater treatment uses microorganisms to degrade organic matter. Aerobic treatment requires oxygen; anaerobic treatment occurs in its absence.

Stages of Anaerobic Digestion (4 Stages): (1) **Hydrolysis:** Complex organic polymers (carbohydrates, proteins, fats) are broken down by extracellular enzymes into simpler monomers — sugars, amino acids, fatty acids, glycerol. (2) **Acidogenesis:** Monomers are fermented by acidogenic bacteria to produce volatile fatty acids (VFAs — propionate, butyrate, acetate), H_2 , and CO_2 . pH drops; this stage is called the "acid phase." (3) **Acetogenesis:** Acetogens convert VFAs to acetate (CH_3COO^-), H_2 , and CO_2 . Syntrophic relationship with methanogens maintains low H_2 partial pressure. (4) **Methanogenesis:** Methanogens (strict anaerobes) convert acetate and H_2/CO_2 to methane (CH_4) and CO_2 — biogas. This is the slowest and rate-limiting step.

Comparison (see table above for detailed comparison): Aerobic processes are faster, produce more sludge, and require energy for aeration. Anaerobic processes are slower to start, produce valuable biogas, generate less sludge, and are more suitable for high-strength (high-BOD) industrial wastewaters.

Advantages of Anaerobic Treatment: (a) Biogas (60–70% CH_4) is a renewable energy source — can power the treatment plant itself. (b) Very low sludge production — reduces sludge disposal costs. (c) No aeration needed — lower operating energy costs. (d) Small footprint for UASB reactors. (e) Digestate can be used as nutrient-rich bio-fertilizer. **Disadvantages:** Long start-up, sensitive to temperature and toxicants, incomplete pathogen removal, potential H_2S odour.

Q7. Explain the treatment scheme for rice mill (agro-based industry) wastewater. How can a zero-waste approach be implemented?

Introduction: Rice milling, especially parboiling operations, generates large volumes of dark brown, high-BOD wastewater (5,000–12,000 mg/L BOD) that can devastate receiving water bodies if discharged without treatment. In India, rice mills are classified as "Red category" industries by CPCB due to their high pollution potential.

Sources of Wastewater: Soaking water (paddy soaked in hot water for parboiling) — richest in organic pollutants; steaming condensate; washing water; mill floor wash-down.

Treatment Scheme: Step 1 — *Screening:* Removal of rice husk, fibrous material, and large solids. Step 2 — *Primary clarification:* Settling of starch granules, bran, and heavy suspended solids in primary sedimentation tank. Step 3 — *Anaerobic treatment (UASB or covered lagoon):* Core treatment stage for high-BOD wastewater. UASB reactor is widely used — upward flowing wastewater passes through a granular sludge blanket; methane produced is captured for energy use. BOD reduced by 70–80%. Step 4 — *Aerobic polishing (ASP or oxidation pond):* Remaining BOD and ammonia oxidised aerobically. Step 5 — *Constructed wetland or slow sand filter:* Low-cost, eco-friendly final polishing before land disposal or irrigation.

Zero-Waste Concept: (1) Biogas from UASB → used for steam generation in parboiling (energy self-sufficiency). (2) Digestate (anaerobic sludge) → bio-fertilizer for rice farms. (3) Rice husk → combustion fuel for steam boiler, or silica extraction for cement. (4) Rice bran → edible bran oil

extraction (high-value product). (5) Treated wastewater → irrigation of paddy fields (N and P recovery). This circular economy approach dramatically reduces environmental impact while improving the mill's economic viability.

7. Quick Revision — Key Facts & Definitions

Term	Key Point
Eutrophication	Excess nutrients → algal bloom → DO depletion → fish kill
AMD	Pyrite + O_2 + H_2O → sulphuric acid; pH 2–4; orange "yellow boy" precipitate
Micro-plastics	Particles <5 mm; nano-plastics <100 nm; ~5 g/person/week ingested
BOD	Biological Oxygen Demand — O_2 needed to oxidise organic matter in 5 days at 20°C
Primary treatment	Physical — screening, grit removal, sedimentation; removes 30–40% BOD
Secondary treatment	Biological (aerobic/anaerobic); removes 85–95% BOD
Tertiary treatment	Advanced — nutrient removal, disinfection, RO; produces near-potable water
Aerobic digestion	With O_2 ; products: CO_2 + H_2O ; high sludge
Anaerobic digestion	Without O_2 ; 4 stages; products: CH_4 + CO_2 (biogas); low sludge
ZLD	Zero Liquid Discharge — no effluent outside plant; mandatory for textiles in India
UASB	Upflow Anaerobic Sludge Blanket — compact anaerobic reactor; used in India
Parboiling effluent	Extremely high BOD (5,000–12,000 mg/L); treated by anaerobic + aerobic
Chromium (tanning)	$Cr(VI)$ — carcinogenic, mutagenic; precipitated as $Cr(OH)_3$ at pH 8–9
Biomagnification	Concentration of pollutants increases up the food chain (e.g., DDT)

Best of luck with your exams! Master the treatment flow diagrams, remember the BOD removal percentages at each stage, and be ready to compare aerobic vs anaerobic processes — these are examiner favourites. ■